RELATIVE MAYER-VIETORIS TRIANGLE FOR LOWER SHRIEK

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Absolute Mayer-Vietoris triangle for lower shriek (more-etale, Lemma lemma-mayer-vietoris-shriek, canonical tag 0GKK). cited at proof line 3880. The triangle on X for the cover U union V = X, pushed forward by Rf_!.
- Composition of lower shriek functors (more-etale, Lemma lemma-shriek-composition, canonical tag 0F7J). cited at proof lines 3881-3882. Identifies Rf_! Rj_{U!} = Ra_! for a = f after j_U (and the V and intersection analogues).
- Rf_! for quasi-finite morphisms (more-etale, Lemma lemma-Rf-shriek-for-quasi-finite, canonical tag 0GKM). cited at proof line 3883. Gives Rj_{U!} = j_{U!} for the quasi-compact open immersion j_U.

Source reference. Original lemma 0GKK: sources/stacks-chow/tags.txt, source lines 20690–20690. Cited at proof line 3880: the absolute triangle applied to K, pushed forward by Rf_!, produces the relative triangle.

Source reference. Original lemma 0F7J: sources/stacks-chow/tags.txt, source lines 19009–19009. Cited at proof lines 3881-3882: identifies Rf_! Rj_{U!} = Ra_! and its V and intersection analogues.

Source reference. Original lemma 0GKM: sources/stacks-chow/tags.txt, source lines 20692–20692. Cited at proof line 3883: rewrites the derived lower shriek of the open immersion.

Lemma 0.2. *Let $f : X \to Y$ be a finite type separated morphism of quasi-compact and quasi-separated schemes. Let U and V be quasi-compact opens of X such that $X = U \cup V$. Denote $a : U \to Y$, $b : V \to Y$ and $c : U \cap V \to Y$ the restrictions of f. Let Λ be a ring. For K in $D^+_{tors}(X_{étale}, \Lambda)$ or $K \in D(X_{étale}, \Lambda)$ if Λ is torsion, we have a distinguished triangle*

$$Rc_!(K|_{U\cap V}) \to Ra_!(K|_U) \oplus Rb_!(K|_V) \to Rf_!K \to Rc_!(K|_{U\cap V})[1]$$

in $D(Y_{étale}, \Lambda)$.

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Proof. This follows from Lemma 0GKK, the fact that $Rf_! \circ Rj_{U!} = Ra_!$ by Lemma 0F7J, and the fact that $Rj_{U!} = j_{U!}$ by Lemma 0GKM. □